INDIAN PEDIATRICS
AI ASSISTANT

STEP 2

Define and Acquire the POC Corpus

A practical guide to issue scoping, article records, file provenance, content roles, exclusions, permission notes, and a repeatable corpus checkpoint.

Teaching manual installment 1.0 | October 2026

Project: Reusable medical-journal literature assistant
POC corpus: Indian Pediatrics, January 2025, Volume 62, Issue 1
Audience: AI students and developers learning to plan and build a grounded literature assistant

How to use this installment
These are the first chapters of a cumulative teaching manual. They explain decisions made so far and provide a reproducible way to understand them. The full 300+ page reference is intended to grow as later roadmap phases are completed.

Document control and alignment

This chapter follows **Phase 2 - Data Acquisition** and the project-approved Step 2 label, **Define and Acquire the POC Corpus**. The selected source is January 2025, Volume 62, Issue 1 of Indian Pediatrics, issue ID 766. The official issue page is https://indianpediatrics.net/articles.php?issue=766. This chapter documents inventory and acquisition only; it does not perform Phase 3 extraction, cleaning or quality correction.

> **Step 2 accepted scope**
> The user approved using the supplied local PDFs for an internal POC and explicitly chose to scope out the two unavailable Correspondence PDFs on pages 81-82. The catalogue preserves those issue entries with a scoped_out status so omission is visible.

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1. What corpus acquisition means

A corpus is the bounded set of documents from which the assistant may retrieve evidence. Corpus acquisition is more than copying files into a folder. A reliable acquisition step answers: which issue, which article records, which local files, which are usable for clinical retrieval, where each came from, what remains absent, and what use basis is recorded.

The publisher issue listing is the catalogue authority for the issue structure. Local PDFs are the acquired file artifacts. Article IDs connect catalogue records to file records now and will later connect extracted text, chunks, embeddings, citations, and evaluation cases. A filename such as `15.pdf` is only a physical filename, not a durable article identity.

1.1 Current source and selected issue

Field	Selected value
Journal	Indian Pediatrics
Issue	January 2025, Volume 62, Issue 1
Publisher issue ID	766
Publication date in project configuration	2025-01-15
Official contents page	https://indianpediatrics.net/articles.php?issue=766
Local corpus folder	Project/data/raw/2025_Jan
Current file count	19 PDF files

2. Scope the issue and define inclusion roles

Before processing files, define what counts as an item and how different materials should be treated. An issue may contain original studies, reviews, clinical case material, invited commentaries, educational articles, clippings, news, notices, corrections, and advertisements. These have different evidentiary value and user intent.

Content role	Default POC behavior	Reason
clinical_evidence	Include in clinical evidence retrieval when article record and text pass later QA.	Primary studies, systematic reviews or guideline summaries may answer clinical-literature questions, but preserve study design and limitations.
clinical_context	May retrieve as contextual material with section and type shown.	Commentaries and perspective pieces are not equivalent to primary research.
nonclinical_context	Catalogue; exclude from clinical answers unless a separate task explicitly calls for it.	Medical education content is relevant to education questions, not treatment evidence.
secondary_source	Retain with reduced ranking and clearly labelled secondary status.	Clippings summarize a separate paper; cite both the clipping and its cited original when available.
administrative_news / notice / advertisement	Catalogue or ignore for clinical evidence; exclude from clinical retrieval.	They are not clinical studies and can produce misleading retrieval matches.
duplicate_or_mixed	Keep in audit manifest; do not index duplicate or administrative portions as article evidence.	Avoid duplicate weighting and contaminated chunks.
scoped_out / unavailable	Retain a catalogue entry with status and reason; no local full text.	Transparent scope is better than silently pretending the issue is complete.

2.1 Inclusion decisions are use-case specific

A commentary may be useful for a question such as 'what perspectives did the journal publish?' while being inappropriate as sole support for a treatment recommendation. A medical education article may be valuable to an educator but not to clinical evidence synthesis. Store stable classifications first, then let each retrieval task define allowed or preferred roles.

3. Article metadata model and retrieval needs

The article catalogue provides an index for both semantic retrieval and deterministic search. Search by author, issue, year, article type, or date should not depend on an embedding model guessing these facts from prose.

Metadata group	Fields in the current catalogue	Retrieval or audit purpose
Identity	article_id, title, journal, issue_id	Stable citations, title lookup, de-duplication and parent-child linking.
People	authors	Author search and disambiguation; blanks mean not yet verified, not 'no authors'.
Publication	publication_date, year, month, volume, issue_number, page_start, page_end	Date ranges, chronology, issue filters and citation display.
Classification	section, content_role	Article type filtering, evidence-aware ranking and role-specific workflows.
Source/provenance	publisher_issue_url, local_filename, access_basis	Trace every record back to source listing and local artifact.
Acquisition/use	acquisition_status, external_demo_permission, notes	Identify available, scoped-out and permission-review state.
Future enrichment	DOI, abstract, keywords, study design, population, topic terms	Improves exact search, evidence filtering and query facets after verification.

3.1 Stable identifier pattern

Current IDs use `IP-2025-V62-I01-###`. Components identify the journal, year, volume, issue, and sequence. Do not derive identity only from a page number: corrections, supplements, multipart articles or renamed files can change page-level conventions. Store the original PDF filename separately.

3.2 Metadata quality rules

- Use the publisher's title and section where available; preserve original capitalization if citation rules need it.
- Keep a field blank if it has not been verified; do not infer authors, DOI, pages or study design from a filename.
- Represent unknown, not applicable, and unavailable as distinct concepts where the data model needs the difference.
- Store raw/source metadata separately from normalized values if normalization can discard meaningful original form.
- Use controlled content-role values consistently, with definitions in the project notes.
- Add topics and keywords only after a documented classification or extraction process; do not turn guesses into authoritative facets.

4. Inventory and provenance of the supplied PDFs

Nineteen physical PDFs were present in `data/raw/2025_Jan`. Before any extraction, the inventory recorded filenames, relative paths, sizes, physical page counts, and whether any selectable text could be extracted. Basic signature and nonempty-file checks reduce the risk of indexing a mislabeled or empty file. These checks do not prove that every page is complete or that OCR quality is good.

4.1 File-to-record mapping

Local file(s)	Mapped item / pages	Treatment in this POC
9.pdf	President's Page, p. 9	Catalogue as context; retrieval eligibility depends on task.
11.pdf	Invited Commentary, pp. 11-12	Context, labelled commentary.
13.pdf	Invited Commentary, pp. 13-14	Context, primary local copy for this article.
14.pdf	Repeated ending from p. 14 plus access notice	Exclude from text ingestion; duplicate/admin mixed file.
15.pdf, 21.pdf, 27.pdf, 31.pdf, 37.pdf, 44.pdf, 49.pdf	Original Articles, pp. 15-52	Clinical evidence candidate; later extraction and QA required.
53.pdf	Research Letter, pp. 53-55	Clinical evidence candidate; preserve article type.
56.pdf	Systematic Review, pp. 56-62; PDF includes unpaginated supplemental material	Clinical evidence candidate; page and supplement boundaries need Phase 3 QA.
63.pdf	Medical Education, pp. 63-70	Nonclinical context; not clinical answer evidence by default.
71.pdf	Update/guideline synopsis, pp. 71-74	Clinical evidence candidate; label as a synopsis of an external guideline.
75.pdf	Reminiscences, pp. 75-77	Historical/context record; not a primary clinical study.
78.pdf	Clinical Case Letter, pp. 78-80	Clinical case evidence; distinguish case-level report from general evidence.
81.pdf and 82.pdf	Two issue-listed Correspondence items	Not supplied; user-approved scoped_out in catalogue.
83.pdf	Clippings: Neonatology	Secondary-source item; limited ranking; preserve cited original study.
84.pdf	News in Brief	Catalogue/audit only; exclude from clinical retrieval.

4.2 Important distinction: physical PDFs versus issue contents

The folder contains 19 PDF files, but that count is not the same as the number of issue entries or the number of documents safe for clinical retrieval. One file repeats content and adds a notice, two issue-listed items are absent, and one supplied file is a news item. The manifest therefore tracks each physical file while the article catalogue tracks editorial entries. Those are separate tables because one source file can be mixed, duplicate, or missing.

5. Classification, inclusion, and exclusions

The catalogue is intentionally broader than the clinical retrieval set. This preserves issue coverage and makes exclusions reviewable without allowing every item to influence a clinical answer.

5.1 Current decisions

- Use 19 catalogue rows for the 19 substantive issue entries represented in this POC catalogue: 16 supplied article records, two issue-listed Correspondence records, and one Clippings record. The two Correspondence entries are explicitly `scoped_out`.
- Track all 19 supplied PDFs in the manifest even where a PDF is nonclinical or excluded.
- Exclude `14.pdf` because its content repeats the end of `13.pdf` and includes an access notice.
- Exclude `84.pdf` News in Brief from clinical evidence retrieval.
- Give `83.pdf` Clippings limited retrieval priority and preserve the cited external paper bibliographic details.
- Keep role labels for commentaries, reminiscences, medical education and cases so the answer layer can distinguish evidence types.
- Record user-provided local copies as the access basis for internal POC. Keep external/editor permission status separately pending confirmation.

5.2 Why not delete excluded files?

An excluded item can still be valuable audit evidence. Keeping a file manifest entry explains what was examined and why it did not enter the clinical retrieval set. The raw file remains unchanged; selection is represented by metadata and later ingestion filters. Deleting source artifacts would make the decision harder to reproduce and might destroy useful provenance.

6. Step 2 implementation walkthrough

The workflow has distinct actions. For another journal, repeat the sequence but adapt the source catalogue and local acquisition method rather than assuming the same URLs, file names, or categories.

Action	What to do	Why / evidence produced
1. Confirm journal and issue	Record journal, publisher URL, issue ID/date/volume/number.	Creates a fixed scope and source citation.
2. Read issue contents	List all sections, articles, pages, authors where available, and non-article content.	Issue completeness can be checked independently of downloads.
3. Set role policy	Define clinical, contextual, secondary, administrative, duplicate and out-of-scope treatment.	Stops the RAG from treating unlike materials as equal evidence.
4. Assign IDs	Allocate stable article IDs once; never use filenames as the canonical key.	Enables parent-child links for text and chunks later.
5. Inventory files	Compare the directory to a manifest; record bytes/pages/text availability.	Detects missing, extra, empty or misidentified files.
6. Record access basis	Record source and user's asserted access/use scope; leave external reuse status explicit.	Prevents a local file from being mistaken for unrestricted permission.

Action	What to do	Why / evidence produced
7. Generate catalogues	Write article metadata and file manifest as UTF-8 CSV with stable column names.	Creates machine-readable inputs for later processing.
8. Run checkpoint	Check schemas, IDs, file match, signatures, text flags, exclusions and scope decisions.	Creates an objective gate before extraction begins.

6.1 Why two CSVs?

`article_metadata.csv` describes the journal's editorial items. `corpus_file_manifest.csv` describes physical local files. One article may be represented by multiple files; one file may contain a duplicate page or notice; and a catalogue entry may not have a supplied file. Combining these facts into one flat sheet encourages duplicate records and ambiguous status. The two tables connect through `article_id` where a relationship exists.

6.2 Preparation utility

`scripts/prepare_step2_metadata.py` is a repeatable generator for the current one-issue catalogue and manifest. Its issue records are explicit source data; its manifest scan checks the live PDF directory and uses PDF parsing to record physical page count and presence of selectable text. If issue scope, schemas, or source metadata changes, update the controlled source records and regenerate the CSVs, then review the diff. Avoid silently hand-editing generated output because regeneration could erase the change.

7. Checkpoint design and recorded result

The checkpoint does not ask whether the final AI assistant is good. It asks whether the selected source set is accurately described and ready to be passed to extraction. The checker is intentionally deterministic and does not call an LLM.

Check	Pass condition	Recorded result
Catalogue schema	Required columns exist and required values are filled.	PASS
Stable IDs	19 unique catalogue IDs are present.	PASS
Physical inventory	Manifest names match exactly the 19 PDFs in the source folder.	PASS
File basics	Each manifest file exists, is nonempty, and begins with a PDF signature.	PASS
Text availability	Selectable text was recorded for every supplied PDF.	PASS; quality and completeness still need Phase 3 review.
Scope gaps	Two unavailable Correspondence entries are explicitly `scoped_out`.	PASS, based on user's decision.
Nonclinical/mixed exclusions	News and duplicate/notice are excluded; clipping is limited.	PASS
Permission reminder	External/editor reuse remains a visible warning.	WARN; not a corpus-structure failure.

7.1 Re-run the checkpoint

From the Project root, with Python available:

```
python scripts\prepare_step2_metadata.py
python scripts\check_step2.py
```

The checker currently reports **9 PASS, 0 FAIL, 1 WARN - READY FOR STEP 3**. The warning does not block internal POC preparation under the user's stated scope; it records that external redistribution/product use requires publisher/editor confirmation. A green Step 2 result means files and decisions are catalogued, not that they are text-clean, legally cleared for any use, or clinically validated.

8. Permissions, risks, and what to avoid

The application must preserve a distinction among public page visibility, user access to a file, permission to download/process it, permission to show it in a demonstration, and permission to redistribute or commercialize it. A URL does not establish all of these. The project records the user's stated local access and intended POC use, while leaving external/editor permission confirmation visible.

Risk / anti-pattern	Why it matters	Control in this project
Assume online availability means public-domain reuse	Access terms and reuse rights are not interchangeable.	Record access basis and external permission separately; seek editor/publisher confirmation for external use.
Index the whole issue without content-role filtering	News, ads, duplicate text or commentary can be misrepresented as clinical evidence.	Use role labels and explicit retrieval eligibility.
Trust filename order	Numbers may be page numbers, IDs, or arbitrary download names.	Use stable article IDs plus a separate filename mapping.
Mark a missing document as present because its catalogue entry exists	The AI could cite a source with no retrieved text.	Use `scoped_out` and no file association; retrieval must operate only on available content.
Treat selectable text as high-quality extraction	Selectable text may still omit columns, tables, figures, symbols or page order.	Phase 3 must inspect and compare extracted samples to rendered pages.
Trust blank metadata as a negative fact	Blank authors/DOI mean unknown or not yet verified.	Keep blanks explicit and add validation status in enrichment.
Publish raw PDFs with the demo code	Local access for a POC is not a redistribution grant.	Keep source PDFs local/ignored by Git and use permitted citation links in the demo.

9. Reuse this process for another journal

A journal onboarding packet should be repeatable. Start from the generic journal config and metadata schema, but let source inspection determine the adapter and categories. Reuse the process, not the assumption that every journal has the same HTML, PDF structure, publication cadence or copyright terms.

9.1 New-journal onboarding template

- Create journal identity/config and record the publisher source pages.
- Document how to navigate decades, years, months, issues, and article detail pages.
- Capture the publisher's category names and map them to the project's controlled content roles.
- Select a small issue with accessible, representative content; record why it is suitable.
- Confirm the basis for collecting and processing local documents before acquisition.
- Create a stable journal-specific ID prefix while keeping the shared metadata columns consistent.
- Build an issue catalogue and manifest; test how mixed, supplemental and missing files are represented.
- Run the Step 2 checks and add journal-specific checks only where the source requires them.
- Write source quirks and decisions in the journal onboarding notes so another student can reproduce them.

10. Handoff into Phase 3

Phase 3 begins only after Step 2 has a bounded set, traceable metadata and explicit file decisions. Its first task is to extract and inspect text, not to create embeddings. Extraction quality is a separate question from whether a PDF opens or has selectable text.

10.1 Phase 3 inputs

- The 19 physical PDFs with unchanged source files.
- The article catalogue with two Correspondence entries scoped out.
- The file manifest with duplicate, News in Brief and Clippings treatment.
- The issue contents page and source URLs.
- Open metadata gaps: most authors and DOI values remain unverified in the catalogue and should be enriched before author-faceted search or final citations.
- The access/use scope note: internal POC as stated by the user; external reuse confirmation remains separately recorded.

10.2 Phase 3 should produce

- One extracted text artifact per selected source/article, with page and article ID provenance.
- A processing log showing method/version and errors.
- A sample inspection report for title, authors, headings, tables, footers, pages and reading order.
- Cleaned text in a derived folder while raw PDFs stay unchanged.
- A quality gate that blocks chunking for corrupt, incomplete or mismatched article text.

References and glossary

The issue categories and article listing should be checked against the publisher's issue page when the corpus is refreshed. The current source is the official January 2025 issue page at https://indianpediatrics.net/articles.php?issue=766.

Glossary

Term	Plain-language meaning
Article catalogue	Structured records about editorial items in the issue.
File manifest	Inventory describing each physical local file and its relationship to an article.
Provenance	Traceable information about where an item came from and how it was handled.
Content role	A controlled label explaining the intended evidence/search role of an item.
Scoped out	Known issue entry deliberately excluded from this POC dataset by scope decision.
Selectable text	Text that a PDF reader can extract; it does not guarantee correct reading order or completeness.
Hybrid retrieval	A search approach combining semantic relevance with exact lexical and metadata matching.